

Alfaytri Maintenance Management System

(MMS)

System Specification & Development Status

Date: June 2026

Version: 2.0

Technology Stack & Infrastructure

The system is built on a modern, enterprise-ready technology stack selected for performance, security, and long-term scalability.

Technology	Purpose	Business Justification
Next.js 15 (React)	Core Framework	Industry-standard web application framework ensuring fast, app-like experience across all devices.
Supabase (PostgreSQL)	Database & Auth	PostgreSQL database with Row-Level Security (RLS) ensuring data protection by user role. Built-in authentication.
GitHub	Version Control	Secure source code management with collaborative development workflows and CI/CD integration.
shadcn/ui & Tailwind CSS	Interface Design	Component library enabling rapid development of a branded, intuitive interface with consistent design system.
TanStack Query v5	Data Management	Client-side data caching ensuring instant dashboard and table loads with background refresh.
Vercel	Cloud Hosting	Zero-configuration deployment infrastructure with automatic scaling, SSL, and edge network.
TypeScript	Type Safety	Full type safety across the codebase reducing runtime errors and improving developer confidence.

Module 1: Contact Centre & Unified Communications

A unified, collapsible sidebar module centralising all customer communications (Voice and Text) alongside CRM data into a single interface. Leverages AI for sentiment analysis and smart geographical dispatching.

1. Unified Inbox & Dialer Navigation

Location: Collapsible Side-Bar

- **3CX Voice Integration:** Built-in UI dialer triggering calls directly from the browser.
 - Backend Routing: All call routing and queuing handled natively by the 3CX server, keeping the web application lightweight.
- **Inbox Navigation Tabs:** Four primary filters: ALL | Unanswered | Tasks | Teams

2. Customer Context Header (CRM Panel)

When a chat or call is active, the sidebar immediately provides the agent with full customer context.

- **Identity:** Customer Name with visual tag (Individual or Business).
- **Contact Info:** Phone Number displayed prominently.
- **Address Management (Draggable Cards):** Agents view and drag addresses into orders.
 - Add Address (Qatar-Specific): Blue Plate system integration for rapid local address entry.
 - Add Address (Global): Standard Google Maps coordinate integration.
- **Installed Products:** Cards for registered services showing Product Name, Installation Date, and Expiration Date.
- **Order History:** Historical Order Numbers and Order Dates for quick reference.

3. Warranty & Replacement Lifecycle Tracking

- Each product maintains a persistent warranty record based on the original installation date, regardless of replacements.
- System calculates: Warranty Start Date, Duration, Expiry Date, and Remaining Warranty (dynamic).
- Warranty replacements do NOT reset the warranty period.
- Linked history chain: Original Installation → Replacement(s) → Follow-ups.
- All warranty replacements logged as Backwork or Follow-up Orders and tagged as Warranty Claims.

4. WhatsApp-Style Chat Thread

- **Omnichannel Inbox:** Consolidates messages from Wati, WHAPI, and 3CX Call Logs into a single chronological thread.
- **Integrated Voice Recordings:** 3CX call history embedded as native voice notes within the thread.
- **Agent Attribution:** Every interaction tagged with the handling agent for accountability.
- **Rich Media:** Message reactions, voice note caching, document/image/video attachments.
- **Message Composer:** Channel switcher (Wati/WHAPI), attachments, canned response templates, emoji picker.

5. Division-Based Channel Segmentation

- Multiple WhatsApp/Wati integrations mapped to specific divisions.
- Dedicated WhatsApp number per division with isolated inbox visibility.
- Agents only view conversations from their authorised division(s).
- Incoming messages auto-tagged and routed by receiving number.
- New divisions can be added without affecting existing channels.

6. AI Sentiment Analysis (Powered by Fanar AI)

Automated customer satisfaction tracking utilising Fanar AI, selected for its Arabic dialect understanding.

- **Text (Wati/WHAPI):** Chat logs scraped, converted to JSON, fed to Fanar AI.
- **Voice (3CX):** Call recordings transcribed and fed to Fanar AI.
- **Positive Sentiment:** Records interaction, attributes to responsible team/agent for recognition.
- **Negative Sentiment:** Flags interaction, escalates to Operations Manager dashboard for proactive resolution.

Module 2: Master Data & System Administration

The central configuration hub for the entire MMS platform. Manages service catalogs, workforce assignments, user permissions, automated notifications, and accounting rules. Broken into 9 configuration areas.

1. Services Hub (/master-data/services)

Central catalog for all operational offerings with 7 management tabs.

- **Normal Services:** N-level tree hierarchy (Category → Service → Sub-service) with pricing, division assignment, duration matrices, and QC items.
- **Contract Services:** Service tree for contracts with contract types, brand linkage, and reliability matrices.
- **Mobile App Services:** Customer-facing service catalog with display hierarchy.
- **Notifications:** Communication templates and recurring triggers with configurable intervals.
- **Instructions:** CRUD manager for field materials (Text/Video/PDF/Images) linked to services.
- **Inventory:** Panel linking products, spare parts, consumables, and tools directly to services.
- **Promotions:** Embedded view of active marketing campaigns and discounts.

2. Teams & Employees (/master-data/teams)

- **Header Controls:** Division filters, global search, grid/list toggle. Quick actions for teams, employees, schedules, and activity logs.
- **Team Cards Grid:** Visual cards showing team name, division, members, vehicle, GPS device status, and tool count.
- **Employee Pool Sidebar:** Staff categorised by availability (Unassigned, Vacation, On-Task, Archived).

3. Users, Roles & Security (/master-data/users)

- **Permissions:** Registry of 79 distinct permission keys, categorised by module.
- **Roles:** Interface to create custom roles by selecting permission combinations.
- **Users:** Directory of all profiles with popups to assign/modify roles and division access.

4. Subscriptions (/master-data/subscriptions)

- **Package Management:** Define Package Name, Price, Duration, Status, and Priority SLA.
- **Actions:** Create, Edit, Archive, and view historical packages.

5. Promotions (/master-data/promotions)

- **Campaigns:** Global rules, eligible divisions, active dates, target discounts.
- **Vouchers:** System-generated codes tied to campaigns.
- **Business Discounts:** Dedicated discounts for entities with Work ID validation.

6. Audit Trail (/master-data/audit-trail)

- **Live Log:** Real-time stream of all system actions across 19 modules.
- **Filters:** Search, Module, Severity Level (Info/Warning/Critical), CSV Download.
- **Deep Dive:** Detailed dialog showing exact JSON diff (old vs. new) and user IP address.

7. Notification Trail (/master-data/notification-trail)

- **History:** Complete log of every sent WhatsApp message and SMS.
- **Filters:** Status (Delivered/Failed), Channel, Trigger Type.

8. Admin Settings (/master-data/admin)

- **Organisation:** Division management (Logos, Stamps, Tax rates, Currency), Warehouse locations, Work Schedules.
- **Catalog & Pricing:** Brand Groups, Pricing Factors, Credit Categories.
- **Operations:** Reason Lists, bilingual Terms & Conditions, Payment Methods.
- **Integrations:** API keys and webhooks for 3CX, GPS (Traccar), Agent Training, WhatsApp division mapping.

9. QuickBooks Connection (/master-data/quickbooks)

- **Account Mappings:** Map internal categories to QB Accounts.
- **Payment Mappings:** Map 8 payment methods to deposit accounts.
- **Item Mappings:** Link MMS Services to QB Items.

Module 3: Orders & Scheduling

The operational core for service fulfilment. Handles the complete lifecycle from quotation to scheduling, execution, backwork, and follow-ups.

1. Order Dashboard (/orders)

- **Smart Filter Chips:** One-click filtering for Scheduled, Pending Approval, No Confirmation, No Address, Past Due.
- **Order List:** Color-coded status badges, division, scheduled date, and confirmation state.
- **Quick Actions:** Manual confirm, rollback confirmations, initiate cancellation workflows.

2. Order Detail & Cancel Dialogs

- **Confirmation Banner:** Visual indicator of 48-hour automated WhatsApp confirmation status.

Tabs: Booked Services | Invoiced & Report | Follow-up & Backwork | Logs

- **Cancellation:** Mandatory reason code with waitlist notification.

3. Create & Schedule Visit (/orders/create)

- **Manual Mode:** N-level service selector, address drag-and-drop from CRM, team calendar with skill highlighting (compatible teams glow green).
- **Mobile Booking (Auto-Assign):** Built-in bookingEngine.ts checks skills, drive times, and conflicts to return best available slots.

4. Quotations (/quotations)

Lifecycle: Draft → Sent → Pending Approval → Customer Approved → Converted (becomes an Order).

- Spare Parts from inventory or one-off Custom Services with manual pricing.
- Threshold-based Submit for Approval routing to managers.
- PDF generation and WhatsApp delivery via Wati/WHAPI.

5. Backwork & Follow-ups

- **Backwork:** Select unfinished services, log reasons, drag onto new calendar slot.
- **Follow-Up:** Flag services requiring return visit with optional additional services.

6. Core Logic & State Management

- **State Machine:** Strictly enforces valid status transitions (e.g. cannot move Cancelled to In-Progress).
- **Inventory Deduction:** FIFO-based automatic stock deduction on order confirmation.
- **Pricing Engines:** Real-time promotion and subscription discount calculation.

Module 4: Contracts & Commercial Agreements

Enterprise-grade system for managing complex, long-term customer agreements across two workflows: Contract Quotations and Live Contracts.

1. Contract Quotations Dashboard (/contracts/quotations)

- **Financial Pipeline:** Live badge showing total QAR value of filtered contracts.
- **Workflow Tracking:** Draft → Manager Review → Awaiting Signature → Active.
- **Quotation Cards:** Total value, monthly breakdown, duration, visit count, areas covered.

2. Create / Edit Contract Quotation

- **Approval & Signature:** Visual progress bar with signed document upload portal.
- **Building Structure Mapping:** Hierarchical site mapping (Complex → Building → Floor → Area) with services attached to specific nodes.
- **Condition Documentation:** Photo/voice media with condition ratings (Good/Fair/Poor) adjusting pricing multiplier.
- **Auto Terms & Conditions:** System crawls building tree and compiles T&C document per service.
- **Payment Schedule:** Fixed (monthly/quarterly), Milestone (percentage splits = 100%), or Completion-based.

3. Live Contracts Dashboard (/contracts)

- **Financial Health:** Outstanding balance tracker across all active contracts.
- **Expandable Cards:** Progress bars for visits and payments; next 6 upcoming visits with team assignments.

4. Contract Execution & Visit Generator

- **Automated Visit Generator:** Set frequency (e.g. bi-weekly), system calculates and plots all visits start-to-end.
- **Batch Order Creation:** One-click bulk creation of scheduled orders linked to the master contract.
- **Team Assignment:** Assign teams to auto-generated visits integrating with the Orders calendar.

Module 5: Invoicing & Payment Processing

Financial ledger managing the complete billing lifecycle from invoice generation to payment collection, credit notes, and QuickBooks synchronisation.

1. Invoice Dashboard (/invoices)

- **Financial Pulse:** Live metrics: Total Outstanding (QAR), Overdue count, Sent, Partial, Paid.
- **Expandable Invoice Cards:** Line-item breakdown, payment history, responsible agent.
- **Voiding:** Authorised void with mandatory reason codes and audit logging.
- **Credit Notes:** Full or partial refunds issued directly against invoices.

2. Payments Ledger (/invoices/payments)

Tracks incoming cash flow across 8 channels: Cash, Bank Transfer, PDC, CDC, Online, Fawran, POS, Pay Later.

- Cheque/transaction tracking with Cheque Number, Date, Bank, and Transaction IDs.

3. Aging Receivables (/invoices/pending-payments)

- **Customer Grouping:** Outstanding debt grouped by customer, sorted highest to lowest.
- **Risk Highlighting:** Red warning border on customers with overdue invoices.
- **Drill-Down:** Popup revealing specific unpaid invoices per customer.

4. QuickBooks Synchronisation

- **Bulk Transfer:** Select multiple items and sync to QuickBooks.
- **State Tracking:** Visual QB badge on synced records preventing duplicates.

5. Customer Payment Portal

- **Self-Service Portal:** Customer-facing read-only invoice view.
- **Online Checkout (Dibsy):** Secure payment link via Edge Function.
- **Atomic Callbacks:** Auto-update invoice balance and issue digital receipt.
- **Multi-Invoice Batch:** Customers select and pay multiple invoices in one transaction.

Module 6: Teams & Workforce Management

Comprehensive workforce and asset allocation engine connecting Back-Office dispatch with the Mobile Team Leader interface.

1. Resource Allocation (/master-data/teams)

- **Split-Screen:** Left Panel (Team Grid) with capabilities; Right Panel (Employee & Vehicle Pool).
- **Drag-and-Drop:** Drag employees into teams, assign Team Leaders, attach vehicles.

2. Live Fleet Tracking (/map)

- **Traccar GPS Integration:** Real-time vehicle and phone tracking on OpenStreetMap.
- **Team Layer:** Live vehicles color-coded: Moving (Green), Idle (Yellow), Stopped (Red), Offline (Grey).
- **Order Layer:** Static pins for scheduled jobs color-coded by status.
- **Quick Search:** Search by team name or plate with FlyTo animation.
- **Geofences:** Draw and manage geofence zones synced with Traccar Cloud.
- **Vehicle History:** Replay historical vehicle trails for any date range.

3. Master Scheduling Board (/calendar)

- **Visual Grid:** Day-view timeline (7 AM to 10 PM) plotting all assignments.
- **Division Chips:** Toggle specific divisions on/off.
- **Visit Type Chips:** Filter by 8 job types, each color-coded.
- **Fit to View:** Entire day on one screen without scrolling.
- **Live Actions:** Swap jobs between teams, edit time slots, dial customers.

4. Mobile Team Leader Interface (/team-leader)

- **Identity Resolution:** Automatically detects logged-in team leader and shows their dispatch list.
- **Daily Run Sheet:** Order cards grouped by date/time, color-coded by job type.

Field Execution: Navigate (Google Maps), Call Customer, Start/In Progress, Customer Not Answering (15-min timer).

Job Completion: Before/After Photos, Inventory Tracking (FIFO), Damage Reports, Invoice Generation.

5. Security & Auditing

- Every action (drag, swap, edit) recorded in master activity log with JSON diffs.

Module 7: Purchase & Sales

End-to-end procurement and sales management covering the full lifecycle from request to payment.

Purchase Module

- **Purchase Orders:** Full lifecycle: RFQ → Draft → Confirmed → Approved, with approval chain and version snapshots.
- **Receivals:** Goods receipt with warehouse assignment and FIFO inventory layers.
- **Bills:** Supplier invoicing with line-item matching to POs.
- **Payments:** Supplier payment tracking across all 8 methods.
- **Shipments:** 17track integration for in-transit shipment tracking.
- **Landed Costs:** Cost allocation across shipment items for true COGS.
- **PO Returns:** Dispatch/cancel/supplier-confirm flow with inventory reversal.
- **Dead Stock Report:** Identifies slow-moving inventory based on configurable thresholds.

Sales Module

- **Sale Orders:** Full lifecycle with customer management.
- **Deliveries:** Picking, packing, and dispatch tracking.
- **Invoices:** Auto-generation from deliveries with PDF output.
- **Credit/Debit Notes:** Refund and adjustment management.
- **Returns:** Customer return processing with inventory restoration.

Development Status Summary

The following table provides a high-level overview of each module and its current development status.

Module	Key Pages	Status
Authentication & Dashboard	Login, Change Password, Dashboard	Completed
Master Data — Inventory	Inventory Items (with category tree)	Completed
Master Data — Suppliers	Supplier management	Completed
Master Data — Users & Roles	Users, Roles, Permissions (79 keys)	Completed
Master Data — Audit Trail	Live audit log with JSON diffs	Completed
Master Data — Admin Settings	Companies, Brand Groups, Credit Groups, Currencies, Warehouses, Approval Settings, Payment Methods, Reason Lists	Completed
Master Data — Services Hub	Normal/Contract/Mobile tree, Notifications, Instructions, Inventory, Promotions, Service Approvals	Completed
Master Data — Teams & Employees	Team cards, employee pool, drag-and-drop, activity log	Completed
Master Data — Service Customers	Customer CRUD, multi-phone, addresses, blacklist	Completed
Purchase Module	Purchase Orders, Receivals, Bills, Payments, Shipments, Landed Costs, Dead Stock, PO Returns, Approvals, RFQs	Completed
Sales Module	Sale Orders, Deliveries, Invoices, Credit Notes, Returns	Completed
Warehouses Hub	7-tab operational hub with stock, movements, adjustments	Completed
Orders Module	Order Dashboard, Create/Edit Order, Team Calendar, Order Detail, Cancel/Confirm/Rollback	Completed
Quotations	Create Quotation, Quotation List, PDF Generation, WhatsApp Send	Completed
Invoices & Payments	View Invoices, View Payments, Pending Payments, Void, Credit Notes	Completed
Contracts — Quotations	Contract Quotation List, Create/Edit Contract Quotation	Completed
Contracts — Live Contracts	Contract Dashboard, Contract Detail, Visit Generator	Completed

Team Leader (Mobile)	Daily Run Sheet, 7 Order Dialogs, Invoice Flow, GPS Tracking, Customer Escalation	Completed
Live Fleet Tracking (Map)	Real-time GPS map, Team/Order layers, Geofences, Vehicle History	Completed
Calendar / Scheduling Board	Day-view grid, Division/Visit Type filters, Swap/Edit	Completed
Payment Portal	Multi-invoice portal, Dibsby checkout, Webhook, Batch Payments	Completed
CSV Import Tool	5 entity type bulk import	Completed
Data Migration	Old system inventory + services import	Completed
Contact Centre — WhatsApp	Chat sidebar, CRM panel, Wati/WHAPI integration, Reactions, Templates	Completed
Contact Centre — 3CX Voice	3CX dialer, call recording playback, voice transcription	Not Started
Contact Centre — AI Sentiment	Fanar AI text/voice analysis, sentiment routing	Not Started
Subscriptions	Subscription packages page (placeholder)	Not Started
Notification Trail	WhatsApp/SMS delivery log	Not Started
QuickBooks Integration	Account/Payment/Item mappings, sync engine	Not Started
Mobile App Services	Customer-facing mobile booking catalog	Not Started
Customer Self-Booking Portal	Self-schedule services within payment portal	Not Started
Reports & Analytics	Overtime report (partial)	Not Started

Summary

Completed Modules	Remaining Modules
24 modules fully developed and operational	8 modules planned for future phases

The platform currently comprises 63 application pages across all modules, with a PostgreSQL database of 120+ tables, Row-Level Security on all tables, and comprehensive audit logging. The system is deployed on Vercel with Supabase as the backend, serving as a fully operational internal ERP for maintenance operations.